

The pace, not the target

The estimand

For each member state: the average annual change in the renewable share of gross final energy consumption, in percentage points per year, estimated as the least-squares slope over the five years ending at the latest observation, and projected to 2030.

Why this and not the target

A target is a statement of intent. A transition plan that assumes the national grid decarbonises on schedule is assuming something the realised data may not support. The pace is observable; the target is not evidence.

The answer

Of 27 member states, 19 fall short of the 42.5% yardstick, 7 reach it, and for 1 the method cannot tell.

That last category exists because the procedure's own out-of-sample error is 5.5 points at a five-year horizon. Any verdict narrower than that would be false precision.

Method

Data

Eurostat nrg_ind_ren, share of renewable energy, balance REN, pulled live from the public dissemination API by src/renewables.py. No key, no manual download, no transcription. Re-running the fetch reproduces the table.

Estimator

Least squares on the window, not the difference between the first and last value. An endpoint difference discards every observation in between and has no variance of its own to report. On Italy the two disagree materially: the endpoint method gives +0.03 points a year, the fitted slope gives +0.38. The first version of this module published the endpoint figure. It was wrong.

The yardstick, stated plainly

42.5% is the EU AGGREGATE target in RED III. It is not a national obligation. It is used here as one common ruler so that member states are comparable. National contributions under the NECPs differ and are not in this dataset. Reading the chart as compliance would be a false premise.

Result

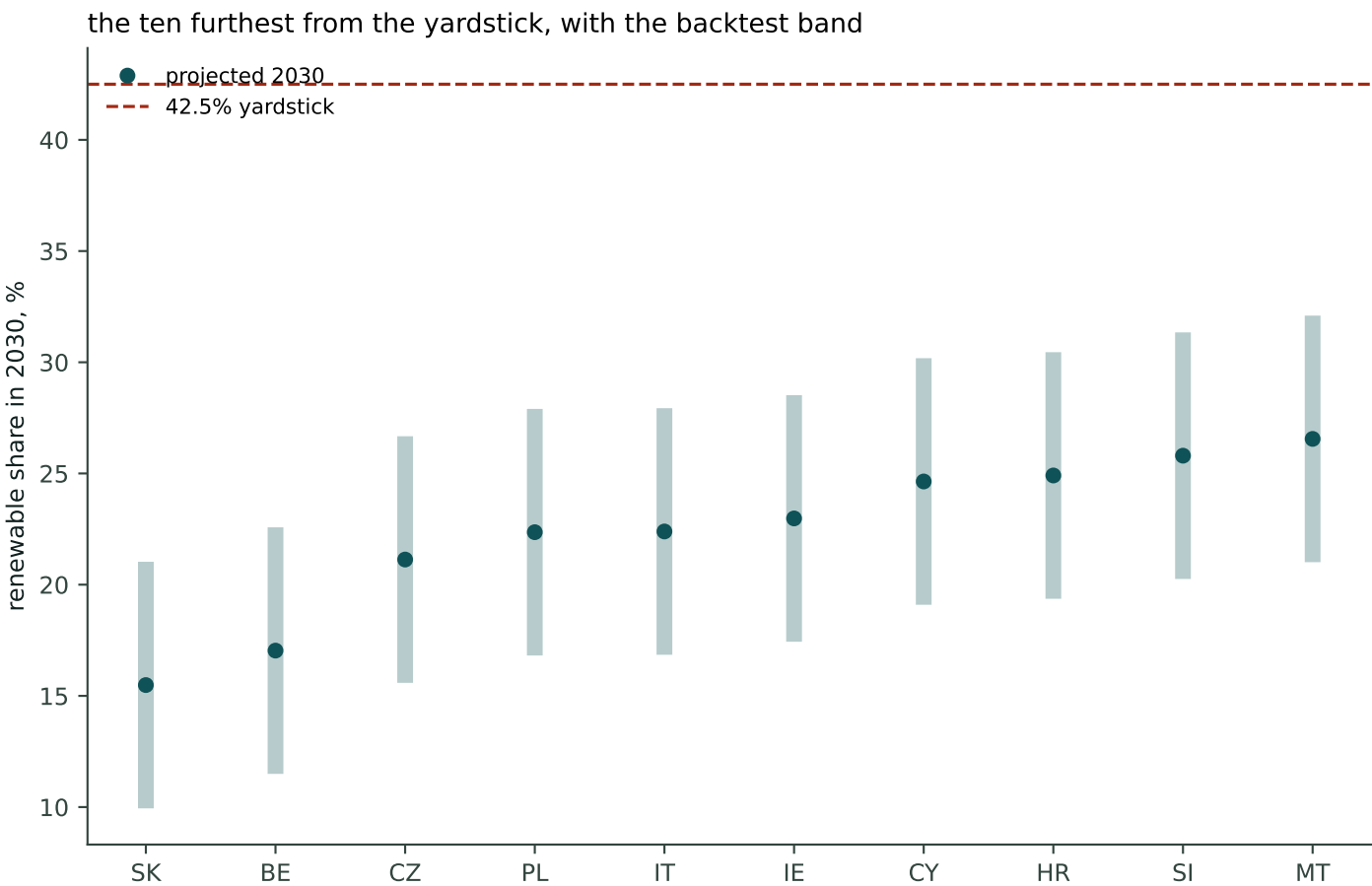
The gap is in the pace, not the level

The ten furthest need between 5.2 and 4.7 points a year. The fastest of them has managed 1.55.

Two that are worth naming

SK: the fitted slope is -0.16 points a year, standard error 0.22. The share is not rising.

IT: +0.38 a year against 4.40 required, from a 20.5% base. Moving, and not nearly fast enough.



Does the method work out of sample?

The test

Fit the same estimator as at 2020, using only data up to that year, predict five years ahead, and compare with what actually happened. This is the only thing that turns a linear projection from an assumption into a number with a measured error.

The result

n = 27 member states. Bias -0.09 points, so the method does not systematically over- or under-shoot. RMSE 5.54 points, mean absolute error 4.18, and 90% of errors within 9.08 points.

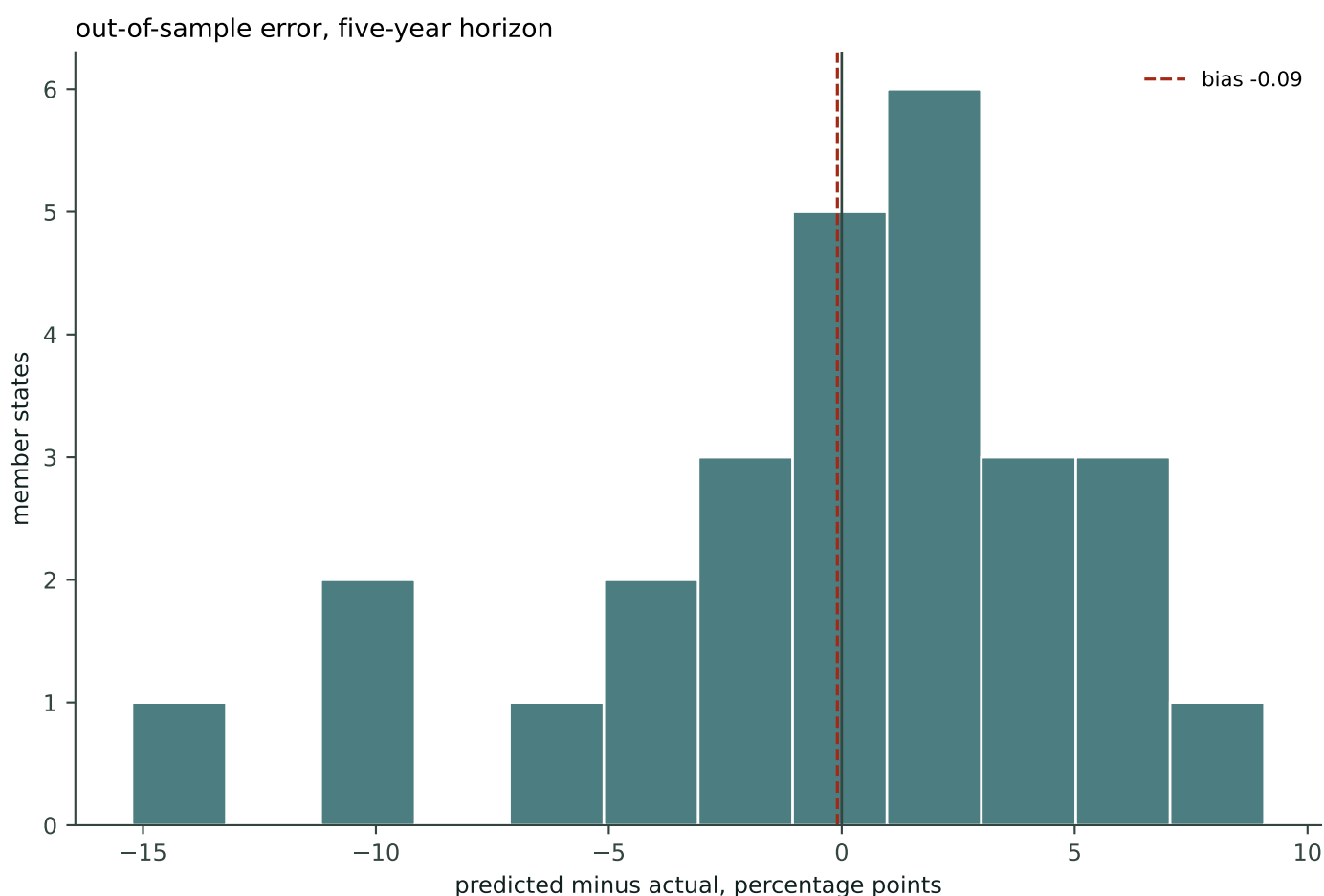
That error is large relative to the gaps on the previous page, and it is why the verdict column refuses to call 1 of 27 either way.

The worst miss

DK: predicted 33.0%, actual

48.2%, error -15.2 points.

Deployment is lumpy, and a single large project moves a small country.



Sensitivity to the arbitrary choice

The choice

The five-year window is not derivable from anything. Three years overweights the 2022 energy shock; ten years reaches back before the directive that changed the incentives. So vary it and see what moves.

The ordering barely moves

Spearman rank correlation of the country ordering against the five-year window:

w = 3 years rho = +0.933

w = 5 years rho = +1.000

w = 7 years rho = +0.964

w = 10 years rho = +0.943

What that buys, and what it does not

The ranking is robust to the window, so the ordering is not an artefact of one choice. The LEVELS are not: the projected 2030 value for a given country moves with the window by more than the gap between neighbouring countries. Use this for who, not for how much.

Limits, and what I would build next

The assumption that breaks it

A constant pace. Deployment is not linear: it responds to auction schedules, grid connection queues and permitting, all of which move in steps. The backtest measures how much that has cost historically, and the answer is a lot.

The denominator

The share is renewables over gross final consumption. A country can gain share by consuming less. This module cannot separate building capacity from shrinking the denominator, and the two have opposite meanings for a transition plan.

The yardstick is an aggregate

Comparing each state against a single EU number is a ruler, not an assessment. The honest version needs the national contributions in the NECPs, which are not machine-readable, and typing them in is exactly what this repository does not do.

Next

Weight each country by gross final consumption and ask whether the EU aggregate reaches 42.5% even when most members do not. That is the number that matters, and it is not the average of these. It needs one further Eurostat series, `nrg_bal_c`, and it is the first thing I would add.